Appendix J - Full Memory Map

The fixed low regions of the Intuition Engine 64-bit physical bus, in address order, with their size and purpose. IE64 can address beyond this low window when RAM is present. IE32, M68K, and x86 see the low 4 GB window directly; the 6502 and Z80 see windowed views into it (Chapters 26 and 27).

J.1 Low RAM and stacks

Range	Size	Purpose
`$000000-$0003FF`	1 KB	M68K vector table; x86 IVT; trap vector base for IE64.
`$000400-$022FFF`	~140 KB	BASIC ROM image, system call shim, monitor stub.
`$023000-$04FFFF`	180 KB	BASIC program text (`BASIC_PROG_START`-).
`$050000-$057FFF`	32 KB	BASIC simple variables.
`$058000-$05FFFF`	32 KB	BASIC string variables.
`$060000-$08BFFF`	176 KB	BASIC arrays.
`$08C000-$08FFFF`	16 KB	BASIC string temporaries.
`$090000-$096FFF`	28 KB	BASIC `GOSUB` / `FOR` stack.
`$097000-$09EFFF`	32 KB	Free RAM (general user).
`$09F000-$09FFFF`	4 KB	IE32 stack (`STACK_START`).

J.2 PC-compatible VRAM apertures

Range	Size	Purpose
`$0A0000-$0AFFFF`	64 KB	VGA graphics window (mode 13h, mode 12h linear).
`$0B0000-$0B7FFF`	32 KB	Reserved (PC-compatible monochrome text window; not implemented).
`$0B8000-$0BFFFF`	32 KB	VGA text buffer (`$B8000`).

J.3 General RAM gap

Range	Size	Purpose
`$0C0000-$0CFFFF`	64 KB	Free RAM (general user).
`$0D0000-$0DFFFF`	64 KB	Voodoo texture RAM.
`$0E0000-$0EFFFF`	64 KB	Free RAM (general user).

J.4 The MMIO region (`$F0000-$FFFFF`)

Range	Size	Device
`$F0000-$F049B`	1180 B	Video chip + palette + extended blitter.
`$F0700-$F07FF`	256 B	Terminal / serial / input.
`$F0800-$F0B7F`	896 B	SoundChip channel registers.
`$F0B80-$F0B91`	18 B	AHX player.
`$F0BC0-$F0BD7`	24 B	MOD player.
`$F0BD8-$F0BF3`	28 B	WAV player.
`$F0C00-$F0C20`	33 B	PSG / AY.
`$F0C30-$F0C3F`	16 B	SN76489.
`$F0C40-$F0CFF`	192 B	SoundChip flex channels 4-6.
`$F0D00-$F0D20`	33 B	POKEY.
`$F0D40-$F0DFF`	192 B	SoundChip flex channels 7-9.
`$F0E00-$F0E1C`	29 B	SID primary registers.
`$F0E20-$F0E2D`	14 B	SID player block.
`$F0E30-$F0E4C`	29 B	SID2 registers.
`$F0E50-$F0E6C`	29 B	SID3 registers.
`$F0E80-$F0EFF`	128 B	SFX triggers.
`$F0F00-$F0F6B`	108 B	TED audio + video.
`$F1000-$F13FF`	1 KB	VGA registers.
`$F1400-$F140F`	16 B	HOST appliance block.
`$F2000-$F2017`	24 B	ULA registers.
`$F2100-$F213F`	64 B	ANTIC.
`$F2140-$F21FB`	188 B	GTIA.
`$F2200-$F221F`	32 B	File I/O.
`$F2260-$F22AF`	80 B	Amiga Paula DMA.
`$F2300-$F231F`	32 B	Media loader.
`$F2320-$F233F`	32 B	RUN loader block.
`$F2340-$F238F`	80 B	Coprocessor.
`$F2390-$F23AF`	32 B	Clipboard bridge.
`$F23B0-$F23BF`	16 B	Coprocessor extended monitor.
`$F23C0-$F23DF`	32 B	IRQ diagnostics.
`$F23E0-$F23FF`	32 B	Bootstrap loader.
`$F2400-$F24FF`	256 B	SysInfo (RAM-size ABI).

Range	Size	Device
`$F8000-$F87FF`	2 KB	Voodoo 3D registers.
`$FA000-$FBAFF`	6912 B	ULA VRAM aperture.

J.5 Main video RAM

Range	Size	Purpose
`$100000-$5FFFFF`	5 MB	Main VRAM aperture for VideoChip framebuffers; large modes may point `VIDEO_FB_BASE` into ordinary RAM.

J.6 Low-window RAM and reserved aliases

Range	Size	Purpose
`$00600000-$FFFEFFFF`	Dynamic	Extended RAM when backed by the current guest-RAM allocation.
`$FFFF0000-$FFFFFFFF`	64 KB	Sign-extended alias of `$00000000-$0000FFFF`.

Guest RAM can extend beyond these fixed low ranges. Use the SysInfo registers in Chapter 23 to discover total and active visible RAM. IE64 can address backed RAM above `$FFFFFFFF`; the compatibility CPUs remain inside the low window or their documented profile caps.

J.7 The 6502 / Z80 views

The 6502 sees a 16-bit address space `$0000-$FFFF`. Its adapter routes the high page-window region as follows:

6502 range	Maps to low bus address
`$0000-$00FF`	zero page in main RAM.
`$0100-$01FF`	6502 stack page in main RAM.
`$0200-$CFFF`	main RAM.
`$D200-$D20A`	POKEY registers.
`$D400-$D40F`	PSG registers.
`$D500-$D55F`	SID registers.
`$D600-$D605`	TED audio registers.
`$D620-$D632`	TED video registers.
`$D700-$D70D`	VGA registers.
`$D800-$D817`	ULA registers, including the paged VRAM data port.
`$E000-$EFFF`	bank-selectable window into main RAM.
`$F000-$FFF9`	MMIO mirror of `$F0000-$F0FF9`, with `$F700-$F705` and `$F7F0` intercepted as bank registers.
`$FFFA-$FFFF`	NMI / reset / IRQ vectors.

The Z80 sees the same shape with port I/O substituted for the chip-aperture region; the bank-window scheme is identical.

J.8 Notes

The map above is the compact engineering view of the low and device regions that have fixed meanings inside the `64`-bit physical bus. Most user programs use a small sub-set of it - the BASIC variable area, the I/O region, and the main video RAM - and never touch the rest. Programs that compete for memory (a large in-memory dataset, a custom blitter routine, a coprocessor worker that wants its own heap) should carve their storage out of the "Free RAM" gaps in J.1, J.3, and J.6, or from boot-sized extended RAM reported by SysInfo, rather than colliding with the BASIC variable region.